K8s-04

1. **Deploy an application using a Deployment with 3 replicas and a rolling update strategy.**

    - Create a nginx file and write yaml script

```
root@Master:~/k8s-04# cd ~/k8s-04
root@Master:~/k8s-04# vi nginx-deployment.yaml
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 3

  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 1

  selector:
    matchLabels:
      app: nginx-deploy

  template:
    metadata:
      labels:
        app: nginx-deploy

    spec:
      containers:
      - name: nginx
        image: nginx:1.25
        ports:
        - containerPort: 80
```

    - Apply and Verify Deployment

```
root@Master:~/k8s-04# kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
root@Master:~/k8s-04# kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   3/3     3            3           13s
```

    - Verify ReplicaSet and Pods

```
root@Master:~/k8s-04# kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
exclude-backend-rs            2         2         2       3d21h
multilabel-rs                 3         3         3       3d21h
nginx-deployment-59cbbdb79d   3         3         3       24s
root@Master:~/k8s-04# kubectl get pods -l app=nginx-deploy
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-59cbbdb79d-f929b   1/1     Running   0          37s
nginx-deployment-59cbbdb79d-k7kj9   1/1     Running   0          37s
nginx-deployment-59cbbdb79d-zk9r9   1/1     Running   0          37s
```

- Verify Rolling Update Strategy

```
root@Master:~/k8s-04# kubectl describe deployment nginx-deployment
Name:                   nginx-deployment
Namespace:              default
CreationTimestamp:      Mon, 29 Jun 2026 10:05:26 +0000
Labels:                 <none>
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=nginx-deploy
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  1 max unavailable, 1 max surge
```

- Rolling update by changing the image from nginx:1.25 to Nginx: latest

```
root@Master:~/k8s-04# kubectl describe deployment nginx-deployment | grep -A5 Strategy
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=nginx-deploy
  Containers:
   nginx:
    Image:        nginx:latest
```

2. Configure a Deployment with a Recreate strategy and observe the downtime.

- Create a file recreate-deployment. Yaml

```
root@Master:~/k8s-04# vi recreate-deployment.yaml
root@Master:~/k8s-04#
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: recreate-deployment
spec:
  replicas: 3

  strategy:
    type: Recreate

  selector:
    matchLabels:
      app: recreate-nginx

  template:
    metadata:
      labels:
        app: recreate-nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.25
        ports:
        - containerPort: 80
```

- Apply and verify the pods

```
root@Master:~/k8s-04# kubectl apply -f recreate-deployment.yaml
deployment.apps/recreate-deployment created
root@Master:~/k8s-04# kubectl get deployment
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment      3/3     3            3           25m
recreate-deployment   3/3     3            3           15s
root@Master:~/k8s-04# kubectl get pods -l app=recreate-nginx
NAME                                   READY   STATUS    RESTARTS   AGE
recreate-deployment-59655859fd-8qj8v   1/1     Running   0          23s
recreate-deployment-59655859fd-gppwb   1/1     Running   0          23s
recreate-deployment-59655859fd-r5hhc   1/1     Running   0          23s
root@Master:~/k8s-04#
```

- Update the Deployment and watch pods

```
^Croot@Master:~/k8s-04#
root@Master:~/k8s-04# kubectl set image deployment/recreate-deployment nginx=nginx:latest
```

```
root@Master:~/k8s-04# kubectl get pods -w
NAME                                  READY   STATUS    RESTARTS      AGE
exclude-backend-rs-22jq6              1/1     Running   1 (37m ago)   3d21h
exclude-backend-rs-pkqmg              1/1     Running   1 (37m ago)   3d21h
multilabel-rs-bv9mk                   1/1     Running   1 (37m ago)   3d21h
multilabel-rs-xm26v                   1/1     Running   1 (37m ago)   3d21h
multilabel-rs-zxr4p                   1/1     Running   1 (37m ago)   3d21h
nginx-deployment-5c8f5c955c-l6cbq     1/1     Running   0             29m
nginx-deployment-5c8f5c955c-p7vgb     1/1     Running   0             29m
nginx-deployment-5c8f5c955c-vmd8m     1/1     Running   0             29m
nginx-nodeport                        1/1     Running   1 (37m ago)   3d22h
nginx-rc-8mqgz                        1/1     Running   1 (37m ago)   3d22h
nginx-rc-j5gsm                        1/1     Running   1 (37m ago)   3d22h
nginx-rc-pxrmg                        1/1     Running   1 (37m ago)   3d22h
nginx-rc-ssfpb                        1/1     Running   1 (37m ago)   3d22h
nginx-rc-z45v9                        1/1     Running   1 (37m ago)   3d22h
recreate-deployment-7c6d446b89-52pcx  1/1     Running   0             2m42s
recreate-deployment-7c6d446b89-l6fc6  1/1     Running   0             2m42s
recreate-deployment-7c6d446b89-xshw4  1/1     Running   0             2m42s
```

- Verify Old ReplicaSet scaled down to 0 and new Replicaset scaled up to 3

```
root@Master:~/k8s-04# kubectl describe deployment recreate-deployment
Name:                recreate-deployment
Namespace:           default
CreationTimestamp:   Mon, 29 Jun 2026 10:30:53 +0000
Labels:              <none>
Annotations:         deployment.kubernetes.io/revision: 2
Selector:            app=recreate-nginx
Replicas:            3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:        Recreate
MinReadySeconds:     0
```

3. Update an existing Deployment and perform a rollback to the previous version.

- Check rollout history

```
root@Master:~/k8s-04# kubectl rollout history deployment nginx-deployment
deployment.apps/nginx-deployment
REVISION  CHANGE-CAUSE
1         <none>
2         <none>
```

- Update the image again

```
root@Master:~/k8s-04# kubectl set image deployment/nginx-deployment nginx=nginx:1.27
deployment.apps/nginx-deployment image updated
root@Master:~/k8s-04# kubectl rollout status deployment/nginx-deployment
deployment "nginx-deployment" successfully rolled out
root@Master:~/k8s-04#
```

- Verify the image and Check rollout history again

```
root@Master:~/k8s-04# kubectl describe deployment nginx-deployment | grep Image:
    Image:         nginx:1.27
root@Master:~/k8s-04# kubectl rollout history deployment nginx-deployment
deployment.apps/nginx-deployment
REVISION  CHANGE-CAUSE
1         <none>
2         <none>
3         <none>
```

- Roll back to the previous version

```
root@Master:~/k8s-04# kubectl rollout undo deployment/nginx-deployment
deployment.apps/nginx-deployment rolled back
root@Master:~/k8s-04# kubectl rollout status deployment/nginx-deployment
deployment "nginx-deployment" successfully rolled out
```

- Verify rollback and rollout history

```
root@Master:~/k8s-04# kubectl describe deployment nginx-deployment | grep Image:
    Image:          nginx:latest
root@Master:~/k8s-04# kubectl rollout history deployment nginx-deployment
deployment.apps/nginx-deployment
REVISION  CHANGE-CAUSE
1         <none>
3         <none>
4         <none>
```

4. **Modify a Deployment to add resource requests and limits for CPU and memory.**

- Edit the YAML and change yaml script and save it

```
root@Master:~/k8s-04# vi nginx-deployment.yaml
root@Master:~/k8s-04#
```

```
  name: nginx-deployment
spec:
  replicas: 3

  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 1

  selector:
    matchLabels:
      app: nginx-deploy

  template:
    metadata:
      labels:
        app: nginx-deploy
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
        resources:
          requests:
            cpu: "250m"
            memory: "128Mi"
          limits:
            cpu: "500m"
            memory: "256Mi"
"nginx-deployment.yaml" 34L, 591B
```

- Apply the updated Deployment

```
root@Master:~/k8s-04# kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment configured
root@Master:~/k8s-04# kubectl rollout status deployment/nginx-deployment
deployment "nginx-deployment" successfully rolled out
```

- Verify the resources

```
root@Master:~/k8s-04# kubectl describe deployment nginx-deployment | grep -A12 "Limits"
    Limits:
      cpu:     500m
      memory:  256Mi
    Requests:
      cpu:        250m
      memory:     128Mi
    Environment:  <none>
    Mounts:       <none>
  Volumes:        <none>
  Node-Selectors: <none>
  Tolerations:    <none>
Conditions:
  Type            Status  Reason
```

5. **Create a Deployment with Max Surge and Max Unavailable configurations.**

- Create surge-deployment. Yaml

```
root@Master:~/k8s-04# vi surge-deployment.yaml
root@Master:~/k8s-04#
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: surge-deployment
spec:
  replicas: 4

  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 2
      maxUnavailable: 1

  selector:
    matchLabels:
      app: surge-nginx

  template:
    metadata:
      labels:
        app: surge-nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
~
```

- Apply it and verify all surge deployment

```
root@Master:~/k8s-04# kubectl apply -f surge-deployment.yaml
deployment.apps/surge-deployment created
root@Master:~/k8s-04# kubectl get deployment surge-deployment
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
surge-deployment   4/4     4            4           11s
root@Master:~/k8s-04# kubectl describe deployment surge-deployment | grep -A5 Strategy
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  1 max unavailable, 2 max surge
Pod Template:
  Labels:  app=surge-nginx
  Containers:
   nginx:
    Image:         nginx
root@Master:~/k8s-04# ^C
```

6. Set up a Deployment with a custom revision history limit.

- Edit the same file: and update revision History Limit to 5

```
root@Master:~/k8s-04# vi surge-deployment.yaml
root@Master:~/k8s-04#
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: surge-deployment

spec:
  replicas: 4
  revisionHistoryLimit: 5

  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 2
      maxUnavailable: 1

  selector:
    matchLabels:
      app: surge-nginx

  template:
    metadata:
      labels:
        app: surge-nginx

    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

- Apply it

```
root@Master:~/k8s-04# kubectl apply -f surge-deployment.yaml
deployment.apps/surge-deployment configured
```

- Verify the revision History Limits

```
root@Master:~/k8s-04# kubectl describe deployment surge-deployment | grep "Revision History"
root@Master:~/k8s-04# kubectl get deployment surge-deployment -o yaml | grep -A2 revisionHistoryLimit
      {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":{},"name":"surge-deployment","na
t":5,"selector":{"matchLabels":{"app":"surge-nginx"}},"strategy":{"rollingUpdate":{"maxSurge":2,"maxUnavaila
ls":{"app":"surge-nginx"}},"spec":{"containers":[{"image":"nginx","name":"nginx","ports":[{"containerPort":8
  creationTimestamp: "2026-06-29T11:08:36Z"
  generation: 2
--
  revisionHistoryLimit: 5
  selector:
    matchLabels:
```

7. **Pause a Deployment during an update, and then resume it.**

- Paused the deployment and check status

```
root@Master:~/k8s-04# kubectl rollout pause deployment surge-deployment
deployment.apps/surge-deployment paused
root@Master:~/k8s-04# kubectl rollout status deployment surge-deployment
deployment "surge-deployment" successfully rolled out
```

- Update the image while paused and check status

```
root@Master:~/k8s-04# kubectl set image deployment/surge-deployment nginx=nginx:latest
deployment.apps/surge-deployment image updated
root@Master:~/k8s-04# kubectl rollout status deployment surge-deployment
Waiting for deployment "surge-deployment" rollout to finish: 0 out of 4 new replicas have been updated...
^Croot@Master:~/k8s-04#
```

- Resume the deployment and check status

```
root@Master:~/k8s-04# kubectl rollout resume deployment surge-deployment
deployment.apps/surge-deployment resumed
root@Master:~/k8s-04# kubectl rollout status deployment surge-deployment
deployment "surge-deployment" successfully rolled out
```

8. **Create a pod using resource requests for memory and CPU, and observe how the scheduler assigns it to a node.**

- Create a yaml file

```
root@Master:~/k8s-04#
root@Master:~/k8s-04# kubectl apply -f scheduler-pod.yaml
pod/scheduler-pod created
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: scheduler-pod
spec:
  containers:
  - name: nginx
    image: nginx
    resources:
      requests:
        cpu: "100m"
        memory: "64Mi"
      limits:
        cpu: "200m"
        memory: "128Mi"
```

- Deleted pod and again created it check status

```
root@Master:~/k8s-04# kubectl delete pod scheduler-pod
pod "scheduler-pod" deleted
root@Master:~/k8s-04# kubectl apply -f scheduler-pod.yaml
pod/scheduler-pod created
root@Master:~/k8s-04# kubectl get pod scheduler-pod -o wide
NAME            READY   STATUS    RESTARTS   AGE   IP                NODE     NOMINATED NODE   READINESS GATES
scheduler-pod   1/1     Running   0          14s   192.168.219.114   master   <none>           <none>
```